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OCTOBER 02, 2025

DANNON YOGURT

Consumer Choice & Demand Analysis

DISCRETE CHOICE MODELING

CONDITIONAL LOGIT

PRICE ELASTICITY

PROC MDC

DATA ARCHITECTURE & SCHEMA

This analysis is built on three linked datasets capturing the full consumer purchase funnel in the Dannon yogurt category, from product attributes and weekly marketing conditions to actual household purchase decisions. Together they enable discrete choice modeling at the household level.

Dataset	Obs	Variables	Role in Analysis
PRODUCTS	6	22	Fixed product attributes — flavor, fat content, package, tier
PRODCHARS	312	6	Time-varying marketing mix — price, display, feature (by store-week)
PANELS	500	4	Household purchase panel — who bought what, when, where

Key Variables

Variable	Dataset	Type	Description
COLUPC	All	Num	Product UPC — links all three datasets
IRI_KEY	PRODCHARS/PANELS	Num	Store/market identifier
WEEK	PRODCHARS/PANELS	Num	Time period (1–52)
price	PRODCHARS	Num	Unit price (USD), varies by week and store
display	PRODCHARS	Num	In-store display indicator (0/1)
feature	PRODCHARS	Num	Feature advertising indicator (0/1)
PANID	PANELS	Num	Unique household identifier
FLAVOR_SCENT	PRODUCTS	Char	Flavor — Strawberry, Blueberry, Peach
FAT_CONTENT	PRODUCTS	Char	Low Fat or Nonfat Light
VOL_EQ	PRODUCTS	Num	Equivalized volume (0.375 pint cup)

PRODUCT SPACE & CHOICE SET CONSTRUCTION

The Dannon yogurt portfolio analyzed consists of 6 unique SKUs differentiated by flavor and fat content. Each SKU was assigned a sequential product ID (JID = 1 through 6) to enable structured discrete choice estimation.

JID	UPC	Flavor	Fat Content	Style	Vol (pint)
1	23663200104	Strawberry	Low Fat	Regular	0.375
2	23663200108	Blueberry	Low Fat	Regular	0.375

3	23663200112	Peach	Low Fat	Regular	0.375
4	33663200620	Peach	Nonfat	Light	0.375
5	33663200622	Blueberry	Nonfat	Light	0.375
6	33663200625	Strawberry	Nonfat	Light	0.375

CHOICE SET ASSEMBLY & DATASET STRUCTURE

To prepare for discrete choice estimation, the weekly product-level choice set was constructed by merging PRODUCTS (fixed attributes + JID) with PRODCHARS (weekly price, display, feature). Five alternative-specific intercepts (J1–J5) were created, with JID=6 as the reference category. This choice set was then merged many-to-many with PANELS using PROC SQL, linking each of the 500 household purchases to all 6 alternatives in that week.

Component	Value	Derivation
Total products (J)	6	Unique UPCs in PRODUCTS dataset
Alternative-specific intercepts	5	J–1, with JID=6 as reference
Household purchases	500	Observations in PANELS
Rows per purchase	6	One row per alternative per decision
Total merged rows	3,000	500 purchases × 6 alternatives
Validation: Sum(bought)	500	Exactly 1 bought=1 per household-week

MARKET CONDITIONS – DESCRIPTIVE ANALYSIS

Overall Dataset (312 store-week-product observations)

Variable	Mean	Std Dev	Min	Max	Interpretation
price	$0.714	$0.124	$0.399	$0.850	Narrow price band — stable category
display	0.019	0.138	0	1	~2% of obs have in-store display
feature	0.019	0.138	0	1	~2% of obs have feature advertising

Price & Promotion by SKU (52 weekly observations per product)

JID	UPC	Avg Price	Display Rate	Feature Rate	Price Range
1	23663200104	$0.728	0.0%	1.9%	$0.40–$0.85
2	23663200108	$0.726	0.0%	1.9%	$0.40–$0.85

3	23663200112	$0.728	1.9%	1.9%	$0.40–$0.85
4	33663200620	$0.704	1.9%	1.9%	$0.40–$0.85
5	33663200622	$0.706	3.8%	1.9%	$0.40–$0.85
6	33663200625	$0.690	5.8%	1.9%	$0.40–$0.85

Key Observation: The yogurt category is price-stable, all SKUs price within a $0.40–$0.85 band with means clustered around $0.70–$0.73. Promotional activity is rare and unevenly distributed. SKU 6 (Nonfat Strawberry) has the highest display rate at 5.8% and the lowest average price, while SKUs 1 and 2 received zero display support. These dynamics suggest that purchase decisions are driven more by habitual preference and occasional in-store visibility than by price cutting.

CONDITIONAL LOGIT MODEL ESTIMATION

Three nested conditional logit models were estimated using PROC MDC in SAS, progressively adding marketing mix variables to the utility function. Each model builds on the previous, allowing direct comparison of fit and parameter stability.

Model I – Intercepts Only (Baseline Preference)

$U_{ij} = \alpha_j + \varepsilon_{ij}$

Parameter	Estimate	Std Error	t-Value	p-Value	Interpretation
J1	-0.7044	0.1843	-3.82	0.0001	Significantly less preferred than J6
J2	-0.1982	0.1579	-1.26	0.2095	Not significantly different from J6
J3	-0.3943	0.1670	-2.36	0.0183	Significantly less preferred
J4	+0.3476	0.1385	2.51	0.0120	Significantly more preferred
J5	+0.1935	0.1432	1.35	0.1765	Slightly positive, not significant

Fit Metric	Value
Log Likelihood	-867.41
Null Log Likelihood	-895.88
LR Statistic	56.95 ($p < 0.01$)
AIC	1,745
BIC	1,766
McFadden Pseudo-R^2	0.032

Model II – Intercepts + Price

$U_{ij} = \alpha_j + \beta \cdot price_{ij} + \varepsilon_{ij}$

Parameter	Estimate	Std Error	t-Value	p-Value	Interpretation
J1	-0.3539	0.1907	-1.86	0.0635	Less preferred, borderline significance
J2	+0.1461	0.1652	0.88	0.3764	Not significantly different from J6
J3	-0.0287	0.1747	-0.16	0.8695	Not different from reference
J4	+0.4096	0.1393	2.94	0.0033	Significantly more preferred
J5	+0.2767	0.1443	1.92	0.0551	Moderately preferred, borderline sig.
price	-7.0021	1.1216	-6.24	<.0001	Strong negative price sensitivity

Fit Metric	Model 1	Model 2	Change
Log Likelihood	-867.41	-841.36	↑ +26.05
AIC	1,745	1,695	↓ –50 (better)
BIC	1,766	1,720	↓ –46 (better)
McFadden R^2	0.032	0.061	Nearly doubled

Model III – Intercepts + Price + Display (Selected Model)

$U_{ij} = \alpha_j + \beta \cdot price_{ij} + \gamma \cdot display_{ij} + \varepsilon_{ij}$

Parameter	Estimate	Std Error	t-Value	p-Value	Interpretation
J1	-0.2674	0.1987	-1.35	0.1782	Less preferred, not significant
J2	+0.2324	0.1742	1.33	0.1824	Not different from reference
J3	+0.0580	0.1834	0.32	0.7517	Not different from reference
J4	+0.4622	0.1431	3.23	0.0012	Significantly more preferred ✔
J5	+0.3319	0.1483	2.24	0.0252	Significantly more preferred ✔
price	-6.9394	1.1240	-6.17	<.0001	Strong negative price sensitivity ✔
display	+0.2791	0.1623	1.72	0.0855	Positive, marginally significant (10%)

Model 3 selected for subsequent analysis. It achieves the lowest AIC (1,694) and the highest likelihood ratio (111.93). While BIC marginally favors Model 2, the display variable is economically meaningful, in-store visibility is a legitimate marketing lever, and its inclusion is theoretically justified. Model 3 balances statistical fit with managerial interpretability.

MODEL COMPARISON – FIT STATISTICS

Model	Specification	Log Likelihood	AIC	BIC	McFadden R^2
Model 1	Intercepts only	-867.41	1,745	1,766	0.032
Model 2	Intercepts + Price	-841.36	1,695	1,720	0.061
Model 3	Intercepts + Price + Display	-839.91	1,694	1,723	0.063

Log Likelihood improves consistently across all three models, confirming each added variable contributes explanatory power. AIC decreases monotonically, favoring Model 3. BIC penalizes additional parameters more heavily, it favors Model 2 but the marginal BIC difference between Models 2 and 3 is only 3 points, well within the range where the economically richer model is preferred.

PRICE ELASTICITY ANALYSIS – OWN & CROSS ELASTICITIES

Using the estimated price coefficient from Model 3 ($\beta = -6.939$) and predicted choice probabilities, own-price and cross-price elasticities were computed for all 6 SKUs. This quantifies how responsive each product's demand is to its own price changes and to price changes in competing products.

Elasticity Formulas:

Own-price elasticity ($j = m$): $E_{jj} = \beta \text{ price} \cdot p_j \cdot (1 - p_j)$

Cross-price elasticity ($j \neq m$): $E_{jm} = -\beta \text{price} \cdot p_m \cdot p_j$

Predicted Market Shares & Own-Price Elasticities

JID	Flavor	Fat Content	Pred. Share	Own Elasticity	Interpretation
1	Strawberry	Low Fat	~10.8%	-4.01	Most price-sensitive
2	Blueberry	Low Fat	~17.8%	-3.75	High price sensitivity
3	Peach	Low Fat	~15.0%	-3.88	High price sensitivity
4	Peach	Nonfat	~22.6%	-3.03	Least price-sensitive — strongest brand
5	Blueberry	Nonfat	~19.7%	-3.19	Moderate sensitivity
6	Strawberry	Nonfat	~14.1%	-3.28	Moderate sensitivity (reference)

Cross-Price Elasticity Matrix

	SKU 1	SKU 2	SKU 3	SKU 4	SKU 5	SKU 6
SKU 1	–4.012	0.630	0.519	0.981	0.840	0.674
SKU 2	0.381	–3.754	0.519	0.981	0.840	0.674
SKU 3	0.381	0.630	–3.885	0.981	0.840	0.674

SKU 4	0.381	0.630	0.519	−3.028	0.840	0.674
SKU 5	0.381	0.630	0.519	0.981	−3.196	0.674
SKU 6	0.381	0.630	0.519	0.981	0.840	−3.279

Diagonal (red) = own-price elasticity. Off-diagonal = cross-price elasticity

All six SKUs exhibit elastic demand, own-price elasticities range from −3.03 (SKU 4, Nonfat Peach) to −4.01 (SKU 1, Low Fat Strawberry). A 1% price increase reduces purchase probability by 3–4% across the portfolio. Cross-price elasticities are consistently positive (0.38–0.98), confirming strong substitution between SKUs. The highest cross-effects (~0.98) involve SKU 4, reflecting its dominant market position, when competitors raise prices, demand shifts most strongly toward SKU 4.

MANAGERIAL RECOMMENDATIONS

The conditional logit model provides a quantitative foundation for SKU-level pricing, promotional, and assortment decisions in the Dannon yogurt portfolio.

Protect SKU 4 & 5 (Nonfat Peach & Blueberry): SKUs 4 and 5 have the strongest intrinsic preference (J4: +0.462, J5: +0.332, both significant). SKU 4 also has the lowest own-price elasticity (−3.03), meaning it is the most defensible against price competition. Prioritize availability, shelf placement, and brand investment for these two SKUs.

Pricing Strategy: All SKUs face elastic demand (elasticity < −1), meaning price cuts increase revenue per unit but compress margins. Price increases should be approached carefully, a 5% price increase on SKU 1 (elasticity −4.01) would reduce purchase probability by ~20%. Consider holding prices stable on high-share SKUs (4, 5) while testing modest increases on lower-preference SKUs (1, 3).

Display Optimization: Display has a positive and marginally significant effect ($\gamma = +0.279$, $p = 0.086$). Currently SKUs 1 and 2 receive zero display support. Allocating even 1–2 display weeks for these SKUs could meaningfully lift trial, particularly for SKU 1 (Low Fat Strawberry), which has the weakest baseline preference and most to gain from in-store visibility.

Cannibalization Risk: Cross-price elasticities are consistently positive and substantial (0.38–0.98). Price promotions on any single SKU will draw volume from the other five, particularly from SKU 4. Portfolio-level promotions should be planned carefully to avoid cannibalizing high-margin SKUs in favor of lower-margin alternatives.

Bottom Line: Price is the dominant lever in this category but it cuts both ways. Elastic demand means promotions drive volume but at margin cost, while price increases risk sharp share loss. The optimal strategy is to invest in display support for under-promoted SKUs, protect the pricing and availability of high-preference SKUs 4 and 5, and manage promotional depth carefully given the strong cross-SKU substitution patterns.

A: APPENDIX – ANNOTATED SAS CODE

All analysis was conducted in SAS using PROC MDC for discrete choice estimation. Code is organized by analytical step.

Data Import & Exploration

```
libname ygt 'path/to/data';


PROC CONTENTS DATA=ygt.products; RUN;

PROC CONTENTS DATA=ygt.prodchars; RUN;

PROC CONTENTS DATA=ygt.panels; RUN;

PROC PRINT DATA=ygt.products (obs=10); RUN;

PROC PRINT DATA=ygt.prodchars (obs=10); RUN;

PROC PRINT DATA=ygt.panels (obs=10); RUN;
```

Count Products & Assign JID

```
PROC SQL;

SELECT count(distinct colupc) as Number_prod FROM ygt.products;

QUIT;


PROC SORT DATA=ygt.products OUT=choice; BY colupc; RUN;


DATA choice_jid;

SET choice;

RETAIN jid 0;

jid + 1;

RUN;
```

Summary Statistics

```
PROC MEANS DATA=ygt.prodchars n mean std min max;

VAR price display feature;

RUN;
```

```
PROC MEANS DATA=ygt.prodchars n mean std min max;

CLASS colupc;

VAR price display feature;

RUN;
```

Build Weekly Choice Set with Intercepts

```
DATA choice_set1;

SET choice_set(keep=jid colupc week price display feature fat_content flavor_scent);

J1=0; J2=0; J3=0; J4=0; J5=0;

if jid=1 then J1=1; if jid=2 then J2=1;

if jid=3 then J3=1; if jid=4 then J4=1;

if jid=5 then J5=1;

LOWFAT=0;

if fat_content='LOW FAT' then LOWFAT=1;

RUN;


/* Many-to-many merge via PROC SQL */

DATA decision_maker;

SET ygt.panels (rename=(colupc=dcolupc));

cid = _N_;

RUN;


PROC SQL;
CREATE TABLE mdc_input AS

SELECT decision_maker.*, choice_set1.*,

(decision_maker.dcolupc=choice_set1.colupc) AS bought

FROM decision_maker LEFT JOIN choice_set1

ON decision_maker.week = choice_set1.week;

QUIT;
```

Model Estimation – PROC MDC

```
/* Model 1: Intercepts only */

PROC MDC DATA=mdc_input;

MODEL bought = J1-J5 / type=clogit choice=(jid 1 2 3 4 5 6);

id cid;

RUN;


/* Model 2: Intercepts + Price */

PROC MDC DATA=mdc_input;

MODEL bought = J1-J5 price / type=clogit choice=(jid 1 2 3 4 5 6);

id cid;

RUN;


/* Model 3: Intercepts + Price + Display (selected) */

PROC MDC DATA=mdc_input;

MODEL bought = J1-J5 price display / type=clogit choice=(jid 1 2 3 4 5 6);

id cid;

RUN;
```

Elasticity Computation

```
/* Extract beta_price from Model 2 estimates */

PROC MDC DATA=mdc_input outest=estimates;

MODEL bought = J1-J5 price / type=clogit choice=(jid 1 2 3 4 5 6);

id cid;

OUTPUT out=pred p=prob;

RUN;


DATA _null_;

SET estimates;

call symputx('beta_price', price);

RUN;
```

```
/* Join all alternatives for elasticity computation */

PROC SQL;

CREATE TABLE elasticity AS

SELECT a.cid, a.jid AS alt_j, b.jid AS alt_m,

a.prob AS p_j, b.prob AS p_m,

a.price AS price_j, b.price AS price_m

FROM probs AS a INNER JOIN probs AS b ON a.cid = b.cid;

QUIT;


DATA elasticities;

SET elasticity;

beta_price = β_price.;

if alt_j = alt_m then

elasticity = beta_price * price_j * (1 - p_j); /* own */

else
elasticity = -beta_price * price_m * p_m; /* cross */

RUN;


/* Average and transpose to matrix */

PROC MEANS DATA=elasticities noprint;

BY alt_j alt_m; VAR elasticity;

OUTPUT out=elas_avg mean=avg_elasticity;

RUN;


PROC TRANSPOSE DATA=elas_avg out=elasticity_matrix(drop=_name_) prefix=alt_;

BY alt_j; id alt_m; VAR avg_elasticity;

RUN;
```

B: APPENDIX – OUTPUT FROM THE ABOVE CODE

Assigning Unique Product IDs (JIDs)

The SAS System

Obs	L1	L2	L3	L4	L5	L9	Level	UPC	SY	GE	VEND	ITEM	_AG_C_1__CATEGORY	VOL_EQ	PRODUCT_TYPE	PACKAGE	FLAVOR_SCENT	FAT_CONTENT	CALORIE_LEVEL	STYLE	ORGANIC_INFO	colupc	jid
1	CATEGORY - YOGURT	RFG YOGURT	GROUPE DANONE	DANNON CO	DANNON	+DANON RFG STB YGRT L/FT 0.375PT	9	00-02-36632-00104	0	2	36632	104	+DANON RFG STB YGRT L/FT 0.375PT 0 2 36632 104 1 1 0.3750RP 02259	0.375	YOGURT	PLASTIC CUP	STRAWBERRY	LOW FAT	MISSING	GRADE	NON ORGANIC	23663200104	1
2	CATEGORY - YOGURT	RFG YOGURT	GROUPE DANONE	DANNON CO	DANNON	+DANON RFG BUB YGRT L/FT 0.375PT	9	00-02-36632-00108	0	2	36632	108	+DANON RFG BUB YGRT L/FT 0.375PT 0 2 36632 108 1 1 0.3750RP 02245	0.375	YOGURT	PLASTIC CUP	BLUEBERRY	LOW FAT	MISSING	GRADE	NON ORGANIC	23663200108	2
3	CATEGORY - YOGURT	RFG YOGURT	GROUPE DANONE	DANNON CO	DANNON	+DANON RFG PCH YGRT L/FT 0.375PT	9	00-02-36632-00112	0	2	36632	112	+DANON RFG PCH YGRT L/FT 0.375PT 0 2 36632 112 1 1 0.3750RP 02254	0.375	YOGURT	PLASTIC CUP	PEACH	LOW FAT	MISSING	GRADE	NON ORGANIC	23663200112	3
4	CATEGORY - YOGURT	RFG YOGURT	GROUPE DANONE	DANNON CO	DANNON	+DNGTF RFG PCH YGRT N/FT 0.375PT	9	00-03-36632-00620	0	3	36632	620	+DNGTF RFG PCH YGRT N/FT 0.375PT 0 3 36632 620 1 1 0.3750RP 02517	0.375	YOGURT	PLASTIC CUP	PEACH	NONFAT	LIGHT	GRADE	NON ORGANIC	33663200620	4
5	CATEGORY - YOGURT	RFG YOGURT	GROUPE DANONE	DANNON CO	DANNON	+DNGTF RFG BUB YGRT N/FT 0.375PT	9	00-03-36632-00622	0	3	36632	622	+DNGTF RFG BUB YGRT N/FT 0.375PT 0 3 36632 622 1 1 0.3750RP 02495	0.375	YOGURT	PLASTIC CUP	BLUEBERRY	NONFAT	LIGHT	GRADE	NON ORGANIC	33663200622	5
6	CATEGORY - YOGURT	RFG YOGURT	GROUPE DANONE	DANNON CO	DANNON	+DNGTF RFG STB YGRT N/FT 0.375PT	9	00-03-36632-00625	0	3	36632	625	+DNGTF RFG STB YGRT N/FT 0.375PT 0 3 36632 625 1 1 0.3750RP 02529	0.375	YOGURT	PLASTIC CUP	STRAWBERRY	NONFAT	LIGHT	GRADE	NON ORGANIC	33663200625	6

Summary Statistics for Product Characteristics

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
price	312	0.7137758	0.1238341	0.3980952	0.8500000
display	312	0.0192308	0.1375558	0	1.0000000
feature	312	0.0192308	0.1375558	0	1.0000000

The MEANS Procedure

colupc	N Obs	Variable	N	Mean	Std Dev	Minimum	Maximum
23663200104	52	price	52	0.7282783	0.1110238	0.3991525	0.8500000
		display	52	0	0	0	0
		feature	52	0.0192308	0.1386750	0	1.0000000
23663200108	52	price	52	0.7263800	0.1128513	0.4000000	0.8500000
		display	52	0	0	0	0
		feature	52	0.0192308	0.1386750	0	1.0000000
23663200112	52	price	52	0.7283168	0.1120108	0.3993865	0.8500000
		display	52	0	0	0	0
		feature	52	0.0192308	0.1386750	0	1.0000000
33663200620	52	price	52	0.7038467	0.1282392	0.3995000	0.8500000
		display	52	0.0192308	0.1386750	0	1.0000000
		feature	52	0.0192308	0.1386750	0	1.0000000
33663200622	52	price	52	0.7061856	0.1348836	0.3980952	0.8500000
		display	52	0.0384615	0.1941839	0	1.0000000
		feature	52	0.0192308	0.1386750	0	1.0000000
33663200625	52	price	52	0.6896474	0.1411414	0.3988889	0.8500000
		display	52	0.0576923	0.2354355	0	1.0000000
		feature	52	0.0192308	0.1386750	0	1.0000000

Creating Intercepts

The SAS System

Obs	FLAVOR_SCENT	FAT_CONTENT	colupc	jid	WEEK	display	feature	price	J1	J2	J3	J4	J5	LOWFAT
1	STRAWBERRY	LOW FAT	23663200104	1	1635	0	0	0.69000	1	0	0	0	0	1
2	BLUEBERRY	LOW FAT	23663200108	2	1635	0	0	0.69000	0	1	0	0	0	1
3	PEACH	LOW FAT	23663200112	3	1635	0	0	0.69000	0	0	1	0	0	1
4	PEACH	NONFAT	33663200620	4	1635	0	0	0.68862	0	0	0	1	0	0
5	BLUEBERRY	NONFAT	33663200622	5	1635	0	0	0.69000	0	0	0	0	1	0
6	STRAWBERRY	NONFAT	33663200625	6	1635	0	0	0.69000	0	0	0	0	0	0
7	STRAWBERRY	LOW FAT	23663200104	1	1636	0	0	0.69000	1	0	0	0	0	1
8	BLUEBERRY	LOW FAT	23663200108	2	1636	0	0	0.69000	0	1	0	0	0	1
9	PEACH	LOW FAT	23663200112	3	1636	0	0	0.69000	0	0	1	0	0	1
10	PEACH	NONFAT	33663200620	4	1636	0	0	0.69000	0	0	0	1	0	0

Assigning Customer-Week Identifier (CID)

The SAS System

Obs	IRI_KEY	PANID	WEEK	dcolupc	cid
1	1085053	3351031	1635	23663200104	1
2	1085053	3135145	1635	23663200108	2
3	1085053	3135145	1635	23663200108	3
4	1085053	3203885	1635	33663200620	4
5	1085053	3336305	1636	23663200108	5
6	1085053	3812628	1636	23663200108	6
7	1085053	3812628	1636	23663200112	7
8	1085053	3134544	1637	23663200104	8
9	1085053	3351031	1637	23663200104	9
10	1085053	3135145	1637	23663200108	10

Model Setup

The SAS System

Obs	cid	WEEK	jid	J1	J2	J3	J4	J5	price	display	feature	bought
1	1	1635	1	1	0	0	0	0	0.69000	0	0	1
2	1	1635	2	0	1	0	0	0	0.69000	0	0	0
3	1	1635	3	0	0	1	0	0	0.69000	0	0	0
4	1	1635	4	0	0	0	1	0	0.68862	0	0	0
5	1	1635	5	0	0	0	0	1	0.69000	0	0	0
6	1	1635	6	0	0	0	0	0	0.69000	0	0	0
7	2	1635	1	1	0	0	0	0	0.69000	0	0	0
8	2	1635	2	0	1	0	0	0	0.69000	0	0	1
9	2	1635	3	0	0	1	0	0	0.69000	0	0	0
10	2	1635	4	0	0	0	1	0	0.68862	0	0	0

Model I

Model Fit Summary	
Dependent Variable	bought
Number of Observations	500
Number of Cases	3000
Log Likelihood	-867.40516
Log Likelihood Null (LogL(0))	-895.87973
Maximum Absolute Gradient	1.57789E-9
Number of Iterations	4
Optimization Method	Newton-Raphson
AIC	1745
Schwarz Criterion	1766

Parameter Estimates

Parameter	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
J1	1	-0.7044	0.1843	-3.82	0.0001
J2	1	-0.1982	0.1579	-1.26	0.2095
J3	1	-0.3943	0.1670	-2.36	0.0183
J4	1	0.3476	0.1385	2.51	0.0120
J5	1	0.1935	0.1432	1.35	0.1765

Model II

Model Fit Summary	
Dependent Variable	bought
Number of Observations	500
Number of Cases	3000
Log Likelihood	-841.36028
Log Likelihood Null (LogL(0))	-895.87973
Maximum Absolute Gradient	2.82908E-6
Number of Iterations	4
Optimization Method	Newton-Raphson
AIC	1695
Schwarz Criterion	1720

Parameter Estimates

Parameter	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
J1	1	-0.3539	0.1907	-1.86	0.0635
J2	1	0.1461	0.1652	0.88	0.3764
J3	1	-0.0287	0.1747	-0.16	0.8695
J4	1	0.4096	0.1393	2.94	0.0033
J5	1	0.2767	0.1443	1.92	0.0551
price	1	-7.0021	1.1216	-6.24	<.0001

Model III

Model Fit Summary	
Dependent Variable	bought
Number of Observations	500
Number of Cases	3000
Log Likelihood	-839.91484
Log Likelihood Null (LogL(0))	-895.87973
Maximum Absolute Gradient	2.8769E-6
Number of Iterations	4
Optimization Method	Newton-Raphson
AIC	1694
Schwarz Criterion	1723

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
J1	1	-0.2674	0.1987	-1.35	0.1782
J2	1	0.2324	0.1742	1.33	0.1824
J3	1	0.0580	0.1834	0.32	0.7517
J4	1	0.4622	0.1431	3.23	0.0012
J5	1	0.3319	0.1483	2.24	0.0252
price	1	-6.9394	1.1240	-6.17	<.0001
display	1	0.2791	0.1623	1.72	0.0855

Elasticity – Own & Cross

The SAS System

Obs	cid	alt_j	alt_m	p_j	p_m	price_j	price_m
1	1	1	1	0.10805	0.10805	0.69	0.69000
2	1	1	2	0.10805	0.17811	0.69	0.69000
3	1	1	3	0.10805	0.14961	0.69	0.69000
4	1	1	4	0.10805	0.22629	0.69	0.68862
5	1	1	5	0.10805	0.19676	0.69	0.69000
6	1	1	6	0.10805	0.14118	0.69	0.69000
7	1	2	1	0.17811	0.10805	0.69	0.69000
8	1	2	2	0.17811	0.17811	0.69	0.69000
9	1	2	3	0.17811	0.14961	0.69	0.69000
10	1	2	4	0.17811	0.22629	0.69	0.68862

alt_j	alt_1	alt_2	alt_3	alt_4	alt_5	alt_6
1	-4.01200	0.63046	0.51930	0.98130	0.84041	0.67398
2	0.38074	-3.75424	0.51930	0.98130	0.84041	0.67398
3	0.38074	0.63046	-3.88488	0.98130	0.84041	0.67398
4	0.38074	0.63046	0.51930	-3.02871	0.84041	0.67398
5	0.38074	0.63046	0.51930	0.98130	-3.19692	0.67398
6	0.38074	0.63046	0.51930	0.98130	0.84041	-3.27887